



Society of Petroleum Engineers

**SPE-227632-MS**

## **Reviving the Production Potential of a Severely Damaged, Subhydrostatic Gas Well Using a Novel Solids-Free Acid Diverter System: Case Study from the Middle East**

Mustansar Raza, A. Pratama, P. Vandeweerd, and A. Almulhim, Halliburton, Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227632-MS](https://doi.org/10.2118/227632-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

---

### **Abstract**

A gas well that produced from a carbonate reservoir developed tubing-casing annulus communication above the polished bore receptacle (PBR), which caused any mechanical means of formation isolation to be impractical before workover operations. The formation was in a subhydrostatic condition after a history of acid fracturing operations, so the operator used a high-volume lost circulation material (LCM) pill to isolate the reservoir.

After the workover, a high-rate stimulation treatment was necessary to reinstate the well's production potential. Subhydrostatic conditions prevented coiled tubing (CT) pinpoint stimulation, so it was decided to bullhead treatment fluid after wellbore cleanout and perforation washing. A diverter system was required, which works independently from the acid system pH and is solids-free, for uniform treatment of the 75-ft interval.

A relative permeability modifier (RPM)-based solids-free acid diverter (SFAD) system was used with 20% diesel-emulsified acid (DEA) and 20% hydrochloric acid (HCl acid) in alternated stages. SFAD changes the effective permeability of the zone, which adheres to the formation and diverts fluid toward untreated zones without reliance on mechanical plugging mechanisms common in typical diverter systems. Elimination of pH-dependent polymers and surfactant-based diverter systems allowed the bullhead treatment of the proposed fluids at a high pump rate.

A specific permeability-based criterion defined the required SFAD volume. The treatment sequence included diverter stages with progressively increased volumes to divert stimulation fluid into zones with higher resistance to treatment fluid.

The bullhead treatment with 20% diesel-emulsified acid, 20% HCl acid, and SFAD increased gas production by 50% compared to preworkover levels. Relative permeability to water-based fluid decreased, which diverted subsequent stages into previously untreated zones. The low-viscosity system adhered instantaneously onto the rock surface, which helped eliminate any requirement for soak time or pH adjustment and proved suitable for high-rate bullhead treatments. Severe subhydrostatic conditions resulted in heavy losses during stimulation, which revealed new insights into SFAD effectiveness.

Typically, chemical diverters used within the industry depend on pH change or a plug-based mechanism. These systems often require tailored soak times, low pump rates, and specific pre- and post-flush or flowback considerations to enable diverter degradation. Moreover, conventional diverters face limitations when used in fractured reservoirs with high permeability contrast. SFAD overcomes these limitations because the system contains no solids, operates without viscosity enhancement for diversion, selectively diverts any water-based system, preserves hydrocarbon zones without damage, and functions effectively in formations up to 6 Darcy permeability.

## Candidate Well

The candidate well was a vertical gas producer completed with 4.5-in. production tubing and 4.5-in. liner to a total depth (TD) of 13,900 ft. The well was completed in tight carbonate and sandstone reservoirs. Fracture stimulation of the sandstone formation was not commercial; therefore, an acid fracture operation was performed in the overlying rock using a plug-and-perforation method. Production results from the carbonate formation met commercial targets. The well was later connected to the production facility.

The well developed a tubing-casing annulus communication because of damage in the production tubing and was subjected to a workover operation. Initial tubing pressure was recorded at 3,000 psi. A viscous fluid was pumped to kill the well, which reduced pressure to only 2,000 psi. The killing operation was revised with a double volume that used 70 bbl of 90-ppb LCM pill, 70 bbl of 170-ppb LCM pill, and 70 bbl of 210-ppb LCM pill, all displaced with viscous brine. A gauge cutter run with a 3.6-in. outer diameter (OD) reached 11,624 ft. A wireline punching tool reached 11,595 ft, followed by a successful tubing punch.

Two barriers—a bridge plug set at 11,545 ft and a freeze plug at surface—were used to nipple down the wellhead after confirmation of plug integrity. The tubing was later cut with an explosive cutter at 11,590 ft, followed by retrieval to surface with a fishing bottomhole assembly (BHA). New tubing was run in hole (RIH), pressure tested to ensure integrity, and drifted with a gauge cutter to a depth of 12,050 ft.

Perforations from 13,085 to 13,160 ft were filled with LCM along with fracture network of the reservoir. The well required revival to reinstate production. [Fig. 1](#) presents the cross section and 3D profile of the well.

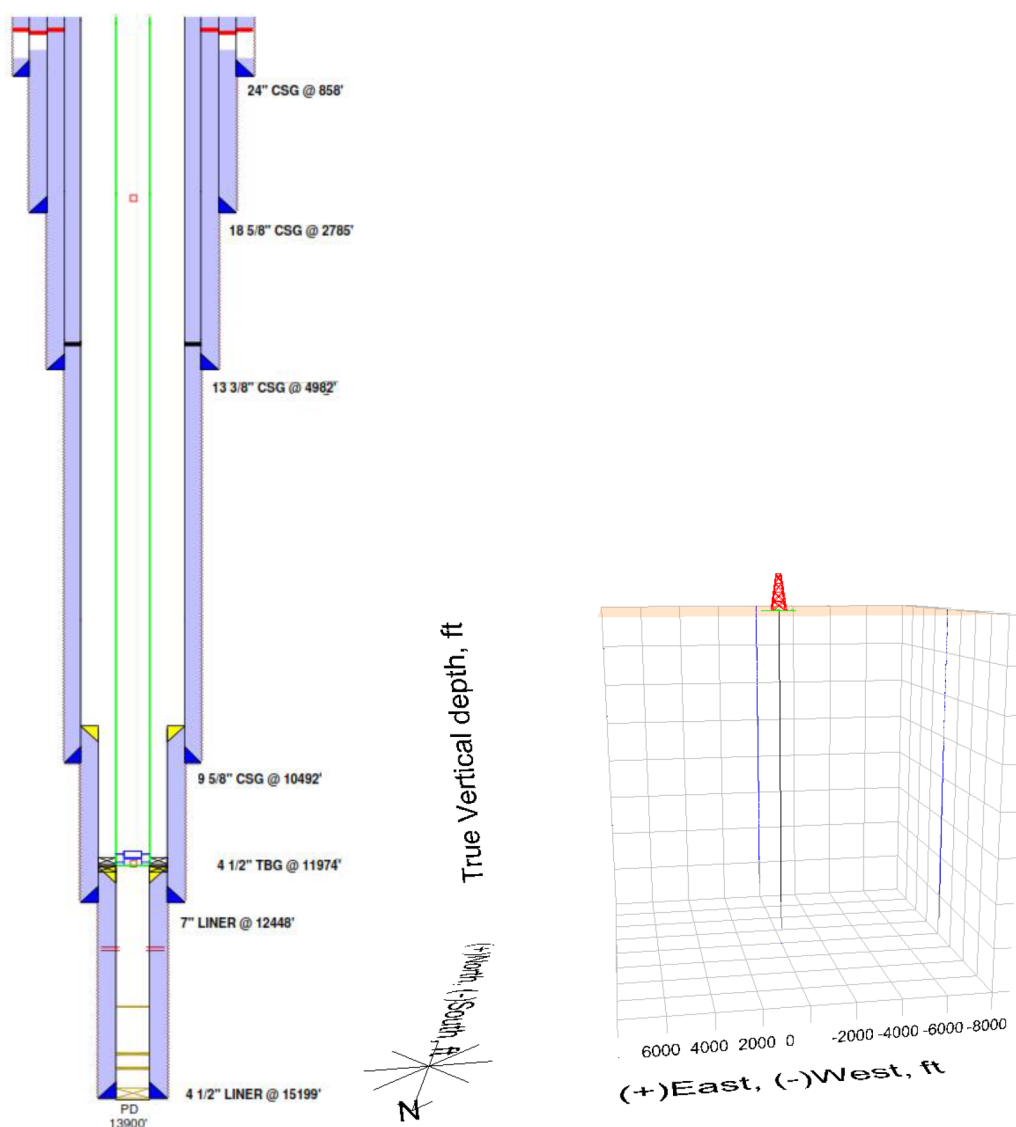


Figure 1—Cross section of the well (left) and inclination (right).

## Carbonate Acid Stimulation

Carbonate acid stimulation serves as the most common method to increase well productivity (Ghommem et al. 2015), typically performed through wellbore cleanout, matrix acid stimulation, and acid fracturing (Singh 1985). Wellbore cleanout removes detritus from well tubulars and perforations that resulted from drilling or deposition of the production fluid during flow periods. This process uses tailored fluid trains combined with mechanical or hydraulic tools.

Matrix acidizing removes the skin from the near-wellbore region through acid injection at a pressure lower than the reservoir's fracture gradient. This technique either etches the rock or dissolves obstructive material within pore spaces, with an eventual aim to reduce the skin value or create negative skin so that hydrocarbons can flow uninterrupted in the wellbore.

Acid fracture stimulation, on the contrary, requires acid pumping at rates that exceed the reservoir fracture pressure. High pressure ruptures the rock, which is subsequently etched by the strong acid as it passes through and forms an irregular pattern. This etched pattern helps prevent full fracture closure upon pressure release.

Factors that impact the performance of matrix acidizing and acid fracturing are a common subject in carbonate acidizing literature. Wellbore cleanout and matrix stimulation is the focus of this paper. Industry guidelines define the requirements of an acid system used in stimulation:

- Treatment fluid must dissolve the rock to create new paths or dissolve damage to reinstate pore space.
- Reactions between treatment fluid, rock, and tubulars should help prevent precipitation to circumvent further damage.
- Treatment fluid should return to surface.
- Placement techniques and control of the treatment fluid should be effective to reach all target zones.

Treatment placement and control are key factors that play a crucial role in carbonate acidizing. Improper diagnosis of well challenges cause stimulation failure, despite progress in software and laboratory capabilities. Proper diagnosis is the first step for the selection of the acid system, volume, and placement technique.

## Acid Systems

HCL acid, organic acids (formic and acetic acid), and certain proprietary blends are the typical acid systems used in carbonate stimulation. The stimulation objective, as determined from the diagnosis, dictates the choice of an acid system. The reaction rate between the acid and reservoir is a decisive factor for acid selection. Fig. 2 shows the summary of reaction data from limestone and dolomite tests performed on 70 core samples from 13 different countries (Gdanski 2005). The graph indicates that the reactivity of dolomite is higher than limestone at a bottomhole temperature (BHT) greater than 200°F.

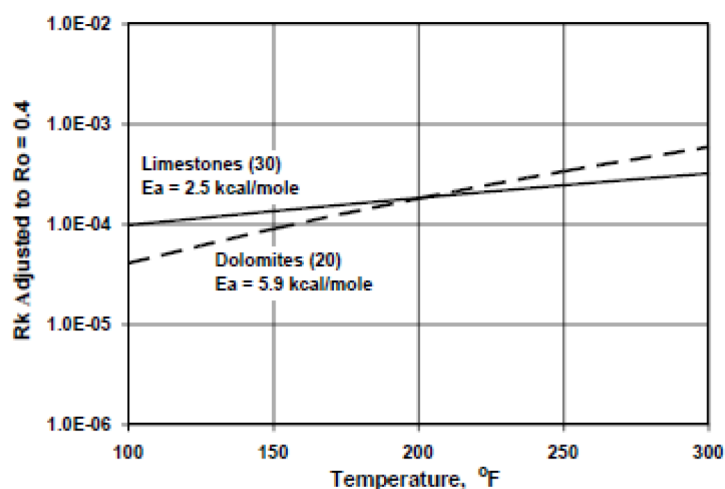


Figure 2—Averaged reactivity of carbonates (Gdanski 2005).

The acid systems are tailored with various additives to help prevent compatibility challenges with reservoir fluid and to protect well tubulars from the corrosive effect of acids.

Poor acid placement and inadequate zonal coverage often cause failure in carbonate acidizing, despite accurate simulations, proper acid selection, and appropriate volume selection (Kalfayan and Martin 2009). This is primarily because acid creates a preferential path when it reacts with the reservoir, which helps prevent the remaining zones from acid contact. Effective diversion techniques ensure acid distribution among all target zones. The following section discusses various diversion techniques and introduces a novel SFAD used in this case study.

## Diverter and Diversion Mechanisms

The importance of acid placement and diverters in matrix stimulation gained industry recognition during the 1970s. [Williams et al. \(1979\)](#) published the use of a rubber packer for zonal isolation to direct acid into desired intervals. The purpose of diversion is to alternate flow paths in zones with similar permeabilities to ensure equal production from all zones ([Gdanski 2005](#)).

Budgetary constraints, best practices established in certain fields, and limitations of simulators to simulate complex reservoirs often discourage the use of diverters. However, advances in diversion technologies and downhole data monitoring have increased industry acceptance. The following sections outline several diversion methods.

### Chemical Diversion

Early diverters were primarily organic materials such as oyster shells, locust bean gum, and soaps. In 1936, soap solutions in HCl acid reacted with calcium chloride to form an oil-soluble, water-insoluble calcium soap ([Harrison 1972](#)). The introduction of HF-based treatments during the 1970s initiated the development of resins, polymer blends, and benzoic acid.

Salt granules were also used as water-soluble diverters but remained insoluble in strong acids, which rendered them ineffective in formations that lacked water production. Benzoic acid flakes and fines, with limited solubility in both oil and water, sublimated to a gaseous state at temperature greater than 230°F, which limited their use in lower-temperature gas wells.

The industry modified fibers originally used for fluid loss and proppant flowback in fracturing applications to serve as diverters. Current fibers consist of dissolvable materials that degrade after stimulation. The use of fibers requires caution during mixing and pumping because it can cause plugging of surface equipment.

Foam formed from surfactant-based fluid and nitrogen emerged as a nonparticulate alternative diverter ([Thompson and Gdanski 1993](#)). Foams have a high viscosity that supports diversion in high-permeability zones with deep damage. Foam diversion performs effectively in high-permeability, low-pressure reservoirs. The high viscosity of foam causes higher friction pressure, which necessitates a detailed assessment of the candidate well conditions before application ([Talabani 1997](#)).

Viscous fluid, such as gelled acids and crosslinked gels, provide controlled leakoff and diversion but can cause formation damage if the fluid fails to flow back ([MaGee et al. 1997](#)) because of the use of a polymer. Viscoelastic surfactant-based gels mitigate this risk through the elimination of polymers and the formation of in-situ gels during stimulation. These surfactants increase viscosity through the formation of micelles that temporarily block high-permeability zones. As flowback begins, viscosity drops, with mutual solvents assisting in the breakdown process ([Chatriwala 2005](#)).

Several studies identify viscous fluids as more effective diverters than particulates. Recent studies support the use of soluble particulates (i.e., fibers and particles) in combination with viscous fluid. These materials create a low-permeability filter cake at the wellbore, redirect flow into tighter zones ([Trif et al. 2024](#)), and dissolve during flowback or after sufficient time.

### Mechanical Diversion

Mechanical methods apply force to isolate zones and allow selective stimulation. Mechanical options include:

- **Bridge Plugs:** Function as temporary or permanent isolation tools for underlying intervals to enable the stimulation of upper zones. Bridge plugs require cylindrical, undamaged pipes for efficient sealing. Inflatable bridge plugs (temporary) and swellable (permanent) bridge plugs can adapt ([Kleverlaan et al. 2005](#)) to irregular surfaces.

- **Packers:** Allow the stimulation of lower zones through the isolation of upper formations. Deployment occurs on either CT or drillpipe. Straddle systems combine packers, plugs, or dual packers to isolate specific zones. Design constraints include differential pressure limits, temperature tolerance, tubing/casing condition, and exposure to CO<sub>2</sub> and H<sub>2</sub>S.
- **Ball Sealers:** First introduced in 1956 (Brown et al. 1963), these balls pump with stimulation fluid to temporarily seal perforations. The sealing redirects flow toward zones with lower intake. Post-treatment, the balls either settle in the rat hole or return to surface, which could compromise the rat hole and require specialized retrieval equipment at the surface. Ball sealers are classified as buoyant (<1 SG) or sinker (>1 SG). Past limitations are compensated with degradable versions that dissolve after a specific period (Bildin et al. 1998). However, ball sealer requirements include an injection string at least three times the ball diameter, a ball OD that exceeds the perforation inner diameter (ID) by 25%, and a minimum injection rate to achieve effective sealing.

### Other Diversion Mechanisms

- **Coiled Tubing:** CT and downhole tools deliver live acid in wells of various depths without the exposure of tubulars to acid. CT ensures precise acid placement at targeted intervals and, when combined with chemical diverters, enhances treatment control. Larger CT sizes (greater than or equal to 2 3/8 in.) accommodate viscous and particulate-laden fluid.
- **Fluid Oscillating Tuned Frequency and Amplitude (FOTFA):** CT tools complement the chemical reaction of acid with hydraulic force. The FOTFA tool uses the Coanda effect—fluid adheres to a curved surface—to generate bursts that alternate (Webb et al. 2006). These bursts create pressure waves that travel through the wellbore without losing kinetic energy, which removes formation damage efficiently. Enhanced permeability allows deeper wave penetration to extend treatment coverage. Acoustic streaming focuses energy near the tool for improved effectiveness (McCulloch et al. 2003).
- **Dynamic Diversion:** Pinpoint stimulation technique that creates hydraulic fractures in deep wellbore sections and microfractures in near-wellbore applications. This mechanism operates under Bernoulli's effect to provide precise fluid diversion (Leguizamon et al. 2022) (Fig. 3) without mechanical isolation. Stagnation pressure, a point of zero fluid velocity and full conversion of kinetic to pressure energy, creates microfractures at specific intervals.
- **High-Rate Diversion:** Maximum pressure differential and injection rate diverts treatment fluid by going above the maximum intake capacity of permeable zones to force fluid into low-intake zones (Paccaloni 1995). This method is effective in depleted carbonate formations to create a network of wormholes for deeper and more uniform acid penetration. Higher injection rates distribute stimulation fluid effectively in all target zones. However, the requirement for significant horsepower creates a commercial challenge for this method.



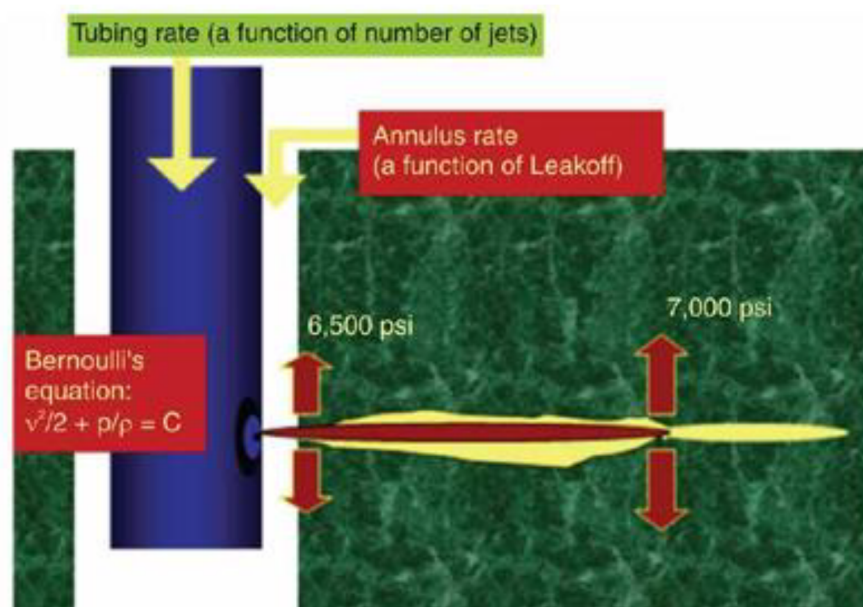


Figure 3—Illustration of dynamic diversion (Al Sebea et al. 2023).

### Solids-Free Acid Diverter (SFAD)

SFAD functions as a solids-free, fluid-loss system that uses a hydrophobically modified polymer originally developed for water control in hydrocarbon-producing wells (Eoff et al. 2004; 2005). Upon contact with the formation, the polymer attaches to the rock surface through electrostatic attraction, which reduces the flow path of aqueous-based fluids (e.g., completion brine, formation water, acid) while hydrocarbon flow remains minimally impacted. This mechanism makes the SFAD an ideal diverter for stimulation candidates with water-bearing zones. Treatment solutions of the SFAD system have low viscosity, typically less than 2 cP. Hydrophobic modification enables the polymer to form multiple layers through interactions with hydrophobic groups (Fig. 4).

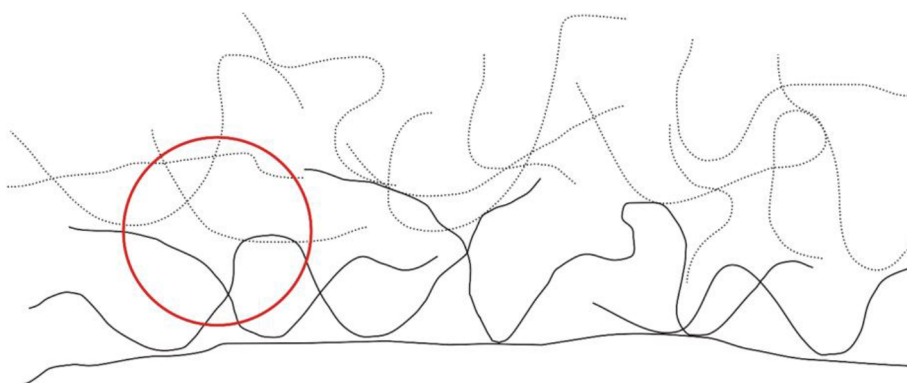


Figure 4—Hydrophobic grouping of the polymer.

The SFAD system effectively reduces water permeability with minimal impact on hydrocarbon permeability (Eoff et al. 2004; 2005). Early laboratory tests showed a rapid pressure increase when the system was injected into cores, which indicated its potential as both a diverter for water-based fluid and a fluid-loss control agent. In a typical test using a carbonate core, the SFAD treatment raised differential pressure from 200 psi to more than 500 psi at a flow rate of 2 mL/min, compared to 10 mL/min for water (Vasquez et al. 2010) (Fig. 5).

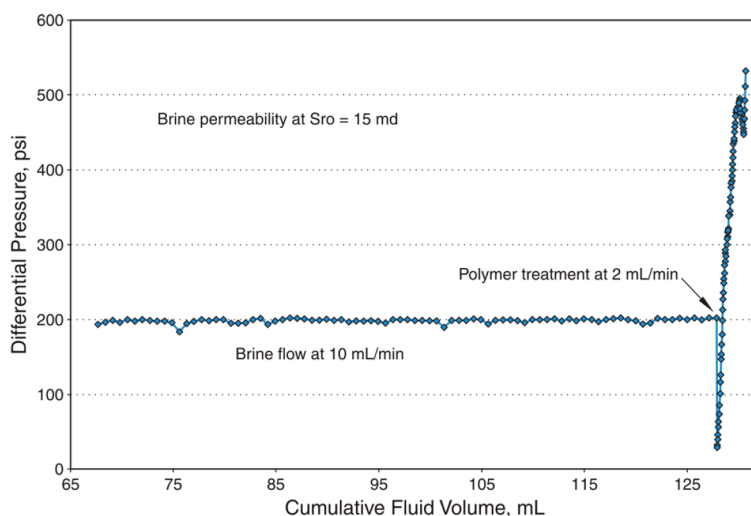


Figure 5—Pressure increments caused by a reduction in relative permeability to water on a limestone core at 175°F.

Fig. 6 illustrates two single-core flow tests. The first core, at residual oil saturation, exhibited a 98% reduction in permeability. The second core, at residual water saturation, exhibited only a 10 to 15% reduction in oil permeability. Test results confirm that SFAD reduces water permeability in water-bearing zones, which helps resist the entry of the acid in the target zone and encourages it to divert toward the hydrocarbon zone that remains unaltered. These characteristics make SFAD a unique and ideal choice for scenarios where water breakthrough is expected.

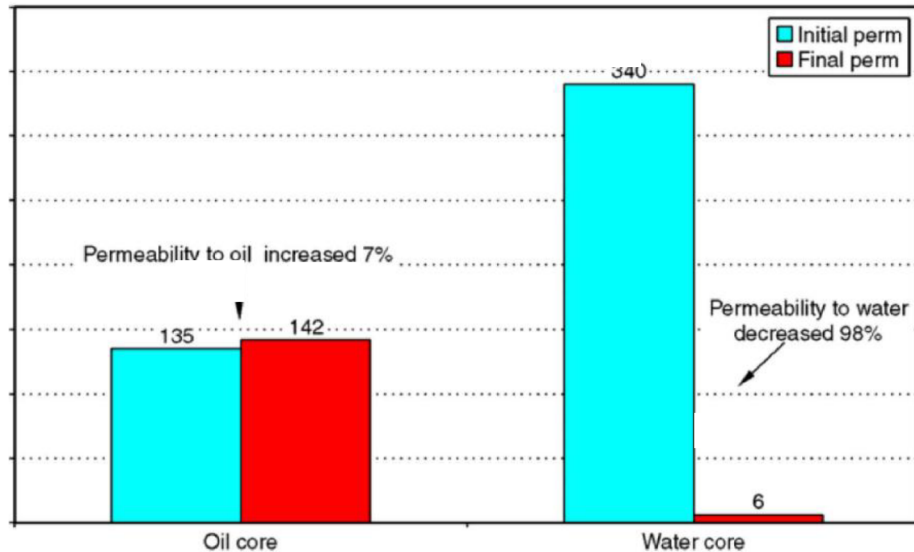


Figure 6—Single-core flow tests with SFAD.

The SFAD system provides several advantages:

- Requires no specialized placement technique
- Pumps through CT or bullheads, based on well conditions
- Adsorbs rapidly onto the rock surface; no shut-in time is required
- Resists acid entirely



- Reduces water permeability more than oil or gas permeability
- Helps prevent stimulation of water-bearing zones; no other diverter in the industry matches this capability
- Mixes easily at the wellsite
- Maintains low viscosity, typically less than 2 to 3 cP

The SFAD system is effective and safe for use in sandstone and limestone formations with a bottomhole static temperature up to 350°F and permeability levels up to 6 Darcys.

## Diagnosis

The carbonate reservoir contained predominantly dolomite lithology with interbedded limestone streaks. The BHT was 270°F, and the pressure gradient equaled 0.357 psi/ft, which indicated subhydrostatic conditions. The well killing operation used a blend of calcium carbonate in various sizes as the LCM. Anticipated compaction of the LCM during workover activities necessitated a mechanical intervention to remove the material from the wellbore.

Subhydrostatic conditions and induced fractures caused LCM loss in the reservoir. The formation damage required a resolution to revive the production potential of the well. The 75-ft perforated interval required analysis of a suitable chemical diversion technique to ensure that the formation damage in the fractured reservoir is properly addressed.

Additionally, a water-bearing layer existed near the wellbore. Water breakthrough has the potential to impact the performance of the well if the stimulation is not designed properly. The initial water cut measured approximately 10%. The objective after the acid fracture was to maintain or reduce the water cut without further stimulation of the water-producing layers.

## Design

The case design included three primary aspects:

- Cleanout to establish access to TD
- Perforations wash with acid
- Stimulation phase to address formation damage

## Cleanout

The obvious approach to clean calcium carbonate-based LCM involves jetting with HCl acid. However, LCM compaction in the wellbore from workover activity, HCl acid exposure to production tubing at high BHT, and neutralized acid returns required special consideration. The cleanout plan included a milling BHA (Fig. 7). The well remained in overbalance conditions (Table 1) and required nitrified cleanout fluid. Conventional positive displacement motors did not meet the requirements. The plan used a metal-to-metal motor with a standard CT BHA for cleanout.

|                                                                |                                                                                    |                                  |                    |                    |                    |                 | Item | Tool Schematic                                                                      |
|----------------------------------------------------------------|------------------------------------------------------------------------------------|----------------------------------|--------------------|--------------------|--------------------|-----------------|------|-------------------------------------------------------------------------------------|
| <h2 style="text-align: center;">Cleanout with Milling BHA</h2> |                                                                                    |                                  |                    |                    |                    |                 | 1    |  |
|                                                                |                                                                                    |                                  |                    |                    |                    |                 | 2    |                                                                                     |
| Item                                                           | Tool Description                                                                   | Tool Connections                 | Tool O/D in Inches | Tool I/D in Inches | T-Length in Inches | Fish neck type  | 3    |                                                                                     |
| 1                                                              | External CT Connector                                                              | 2 3/8" PAC PIN                   | 3.13               | 1.38               | 9                  |                 | 4    |                                                                                     |
| 2                                                              | MHA Consisting of:<br>Check Valve<br>Disconnect<br>Circ Sub<br>Burst Disc: 7.5kpsi | 2 3/8" PAC Box<br>2 3/8" PAC Pin | 2.88               | 0.75               | 52                 | GS Pulling Tool |      |                                                                                     |
| 3                                                              | Metal-To-Metal Motor                                                               | 2 3/8" PAC Box<br>2 3/8" PAC Pin | 2.88               | N/A                | 162                |                 |      |                                                                                     |
| 4                                                              | 5 Bladed Concave Junk Mill                                                         | 2 3/8" PAC Box                   | 3.560              | 1.2                | 14                 |                 |      |                                                                                     |
| Maximum Tool-String OD:                                        |                                                                                    |                                  | 3.56               |                    |                    |                 |      |                                                                                     |
| Minimum Tool-String ID:                                        |                                                                                    |                                  |                    | 0.75               |                    |                 |      |                                                                                     |
| Total Length of BHA in Feet:                                   |                                                                                    |                                  |                    |                    | 20.43              |                 |      |                                                                                     |

Figure 7—Cleanout the BHA with a metal-to-metal motor.

Table 1—Hydrostatic head with reservoir pressure had overbalance conditions.

|                                  |                            |
|----------------------------------|----------------------------|
| Reservoir Pressure               | 4,750 psi at 13,160 ft TVD |
| Hydrostatic pressure w/ 8.34 ppg | 5,760 psi                  |
| Overbalance Conditions           | 1,010 psi                  |

Cleanout simulations used available LCM data. The cleanout hydraulics in Fig. 8 indicated a requirement of 1,000 scfm of nitrogen with 1.6 bbl/min of liquid to achieve a velocity factor of 10 using a 2 3/8-in. CT string under loss conditions.

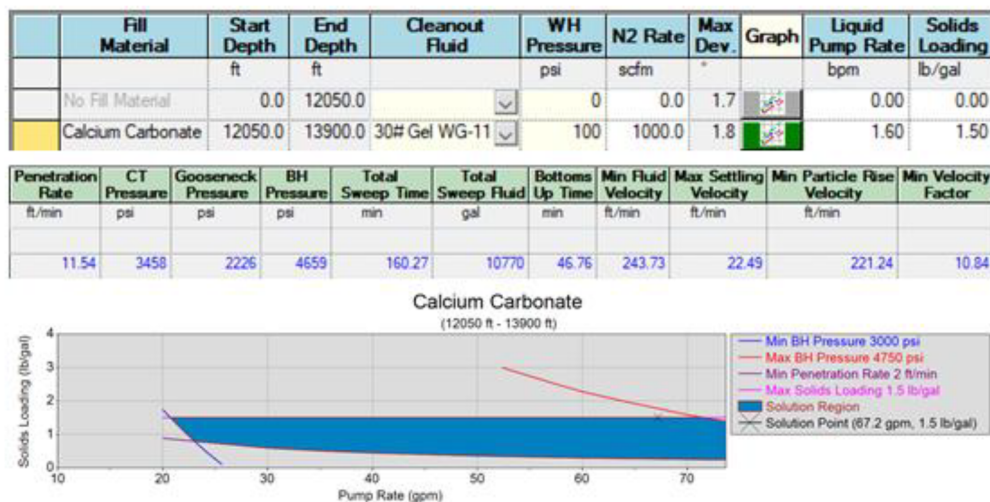


Figure 8—Calcium carbonate cleanout simulations.

Accordingly, the cleanout schedule included treated water (TW) and 30# gel (Table 2). Slickline tagged the top of LCM at 12,050 ft. The cleanout schedule was designed to start at the same depth. Batches of 20% HCl acid were pumped every 1,000 ft of penetration to remove potential LCM adhered to the production tubing walls. Gel sweeps were pumped every 250 ft to lift the LCM to surface, chased by TW.

Table 2—Cleanout schedule of the LCM.

| Stage       | Direction | Total Distance (ft) | CT Speed (ft/min) | Fluid Rate (bbl/min) | N2 Rate (scfm) | Stage Time (min) | Fluid Volume (bbl) | Fluid at Nozzle | WHP (psi)   |
|-------------|-----------|---------------------|-------------------|----------------------|----------------|------------------|--------------------|-----------------|-------------|
| Bite        | RIH       | 250                 | 10                | 1.6                  | 1000           | 25               | 40.0               | TW              | ~300        |
| Circulation | Station   | 0                   | 0                 | 1.6                  | 1000           | 7                | 11                 | 30# Gel         | Overbalance |

> Repeat the step until CT reaches plug-back TD.

> Perform pull test every 500 ft of penetration.

> Pump 10 bbl of 20% HCl acid after every 1,000 ft of cleanout.

> Once at TD, pump 20 bbl of 30# gel, 20 bbl of TW, and 20 bbl of 30# gel, followed by 1.5× bottoms up with TW.

## Perforation Wash

Perforations were expected to be plugged with LCM, and the cleanout was designed with FOTFA BHA (Fig. 9). The acid wash was designed with 20% HCl acid to complete one cycle in the perforations to ensure full access for acid intake during the stimulation operation. The perforation wash design called for a shut-in well; therefore, the plan included an injection test before the acid wash.

| Perforation Wash |                                                                                    |                                  |                    |                    |                    |                 | Item | Tool Schematic                                                                        |
|------------------|------------------------------------------------------------------------------------|----------------------------------|--------------------|--------------------|--------------------|-----------------|------|---------------------------------------------------------------------------------------|
|                  |                                                                                    |                                  |                    |                    |                    |                 | 1    |  |
|                  |                                                                                    |                                  |                    |                    |                    |                 | 2    |                                                                                       |
|                  |                                                                                    |                                  |                    |                    |                    |                 | 3    |                                                                                       |
|                  |                                                                                    |                                  |                    |                    |                    |                 | 4    |                                                                                       |
| Item             | Tool Description                                                                   | Tool Connections                 | Tool O/D in Inches | Tool I/D in Inches | T-Length in Inches | Fish neck type  |      |                                                                                       |
| 1                | External CT Connector                                                              | 2 3/8" PAC PIN                   | 3.13               | 1.38               | 9                  |                 |      |                                                                                       |
| 2                | MHA Consisting of:<br>Check Valve<br>Disconnect<br>Circ Sub<br>Burst Disc: 7.5kpsi | 2 3/8" PAC Box<br>2 3/8" PAC Pin | 2.88               | 0.75               | 36                 | GS Pulling Tool |      |                                                                                       |
| 3                | Straight Bar                                                                       | 2 3/8" PAC Box<br>2 3/8" PAC Pin | 2.88               | N/A                | 48                 |                 |      |                                                                                       |
| 4                | Jetting Nozzle                                                                     | 2 3/8" PAC Box<br>2 3/8" PAC Pin | 2.88               | N/A                | 9                  |                 |      |                                                                                       |
|                  |                                                                                    | Maximum Tool-String OD:          | 3.125              |                    |                    |                 |      |                                                                                       |
|                  |                                                                                    | Minimum Tool-String ID:          |                    | 0.75               |                    |                 |      |                                                                                       |
|                  |                                                                                    | Total Length of BHA in Feet:     |                    |                    | 8.50               |                 |      |                                                                                       |

Figure 9—BHA design for perforation wash.

## Stimulation

The well's damage mechanism was identified. A proprietary simulator determined the treatment volume, diversion mechanism, and skin reduction. Field-specific experience and best practices supported the design. The LCM and mud formulations provided input for the Formation Damage Advisor. The advisor ranked calcium carbonate and solids plugging as the most severe damage mechanisms (Fig. 10).

| Damage Name                  | Score                  | Score % | Selected                            |
|------------------------------|------------------------|---------|-------------------------------------|
| CaCO <sub>3</sub> Deposition | <div><div></div></div> | 19.91   | <input checked="" type="checkbox"/> |
| Solids/Plugging              | <div><div></div></div> | 19.45   | <input checked="" type="checkbox"/> |
| CaSO <sub>4</sub> Deposition | <div><div></div></div> | 15.93   | <input type="checkbox"/>            |
| SrSO <sub>4</sub> Deposition | <div><div></div></div> | 11.20   | <input type="checkbox"/>            |

Figure 10—Formation Damage Advisor.

Acid volume, pumping rate, and the diversion mechanism to reinstate fracture conductivity were simulated. The water-bearing layer and fracture permeability required a diverter that allowed diversion while live acid access to the water-bearing zone was minimized. Best practices and software recommendations identified gelled acid, foamed fluid, or ball sealers as suitable options for the diversion mechanism (Fig. 11). However, these mechanisms lacked control over live acid entry into the water-bearing zone. Therefore, simulations were performed with SFAD to ensure that effective diversion is achieved for the uniform treatment of the net perforation interval, while helping prevent the stimulation of the water-bearing zone.

| Method Name          | Score                  | Score (%) |
|----------------------|------------------------|-----------|
| Ball Sealers         | <div><div></div></div> | 77.64     |
| Foams                | <div><div></div></div> | 14.91     |
| Gels and Gelled Acid | <div><div></div></div> | 5.96      |

Figure 11—Diversion Advisor.

The effectiveness of the diverter was confirmed through fluid placement graphs to ensure that acid penetration is sufficient to bypass the fracture damage in the perforated interval. Subhydrostatic conditions and fractured permeability were expected to induce the suction behavior of the well, as indicated in historical data recorded before the acid fracture operation. Fluid placement simulations with SFAD indicated sufficient diversion to meet the minimum acid penetration requirement (Fig. 12).

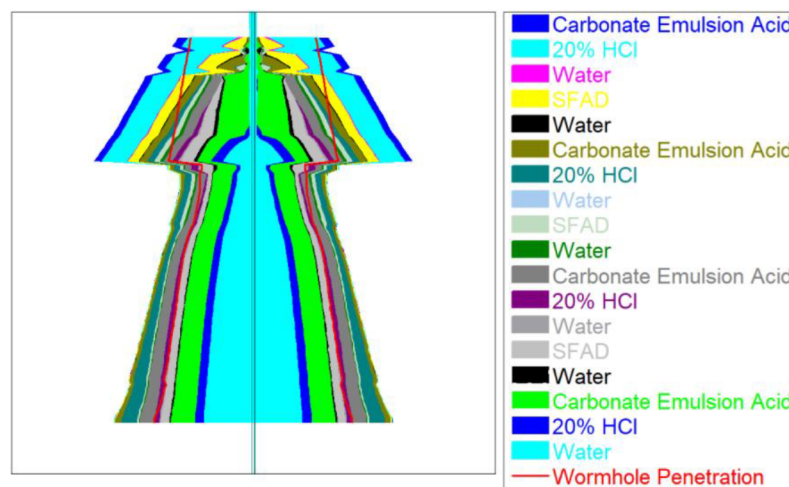


Figure 12—Treatment placement simulation without diverter.

Permeability and temperature criteria determine diverter concentration and volume. Fundamental design highlights include:

- **Main Acid:** The solubility of dolomite is greater than limestone at BHT greater than 200°F; therefore, the treatment used 20% HCl acid. Because of the rapid reactivity of HCl acid in temperature greater than 250°F, HCl acid alone did not provide sufficient acid penetration in the reservoir. Expected damage existed near the wellbore and deep within fractures, which required a dual-acid approach. DEA, with its retarded reaction profile, was designed in stages to address the deep wellbore requirements, while 20% HCl acid addressed near-wellbore requirements. The acid treatment concentration was set at 169 gal/ft to target the intended penetration in the reservoir based on simulation outcomes.
- **Diverter:** SFAD concentration was set at 33 gal/ft using progressively larger volumes after each acid stage. Unlike conventional diverters that pumped in equal volumes after each stage, SFAD volumes increased incrementally. A small amount (9 bbl) was pumped after the first acid batches because of expected initial fracture plugging. As the acid created a low-resistance path in the reservoir, a higher concentration of the RPM was required to resist the entry of subsequent water-based fluid. Therefore, the second stage of the diverter was set at 21 bbl, followed by 30 bbl before the final acid stages. The simulation supported the effectiveness of this incremental increase in diverter volume.
- **Conveyance Medium:** The well's suction behavior and high-rate requirement necessitated stimulation pumping in a conventional bullhead manner.

## Execution

### Cleanout Run

CT was RIH to the slickline-tagged depth with the milling BHA. The CT weight failed to indicate the presence of LCM at the anticipated depth; therefore, further RIH was performed. Surface returns appeared without nitrified treatment fluid initially because of the blocked perforations. Once the cleanout progressed and perforation clearance was established, losses began. Cleanout continued with nitrified TW and 30# gel as designed.

Tubing pickling was performed with 20% HCl acid after every 750 ft. Gel batches and bottoms-up circulation followed once CT created TD (Fig. 13). CT was retrieved to the surface after clear returns were observed.

Wellhead pressure dropped while CT was pulled out of hole (POOH) to the surface, which confirmed communication with the fracture network.

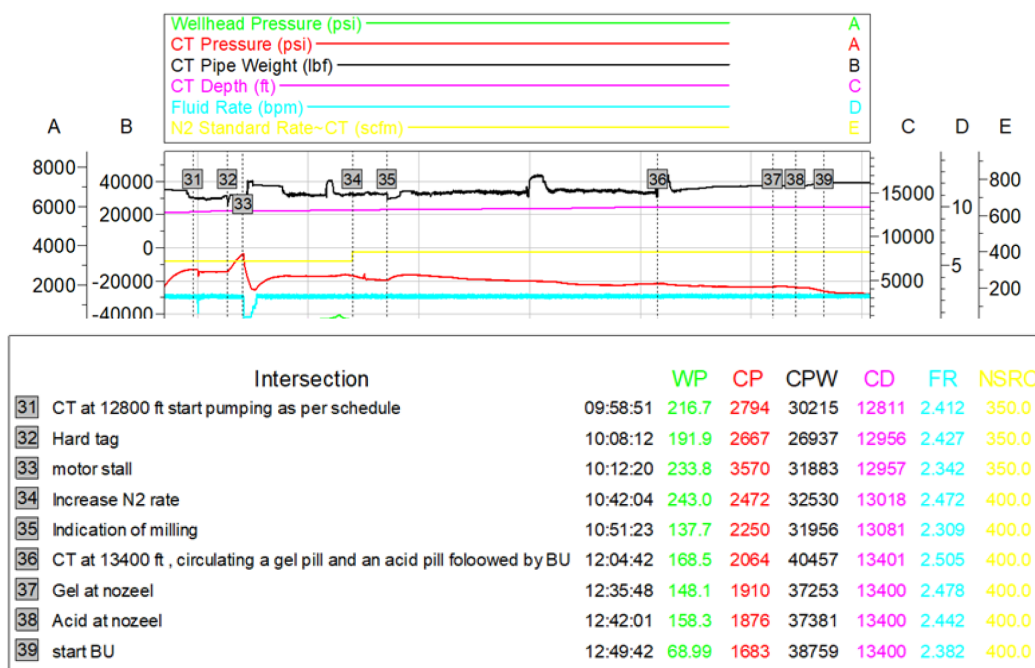


Figure 13—Cleanout with milling BHA.

### Acid Wash

CT reached the perforation depth with side-jetting FOTFA. A step-rate injection test was performed through CT and CT-tubing annulus. The wellhead pressure remained negative at a maximum pumping rate of 5 bbl/min.

The acid wash was performed through the completion of one cycle in the perforations with the well closed from the surface (Fig. 14). CT was POOH to surface to proceed with the stimulation.

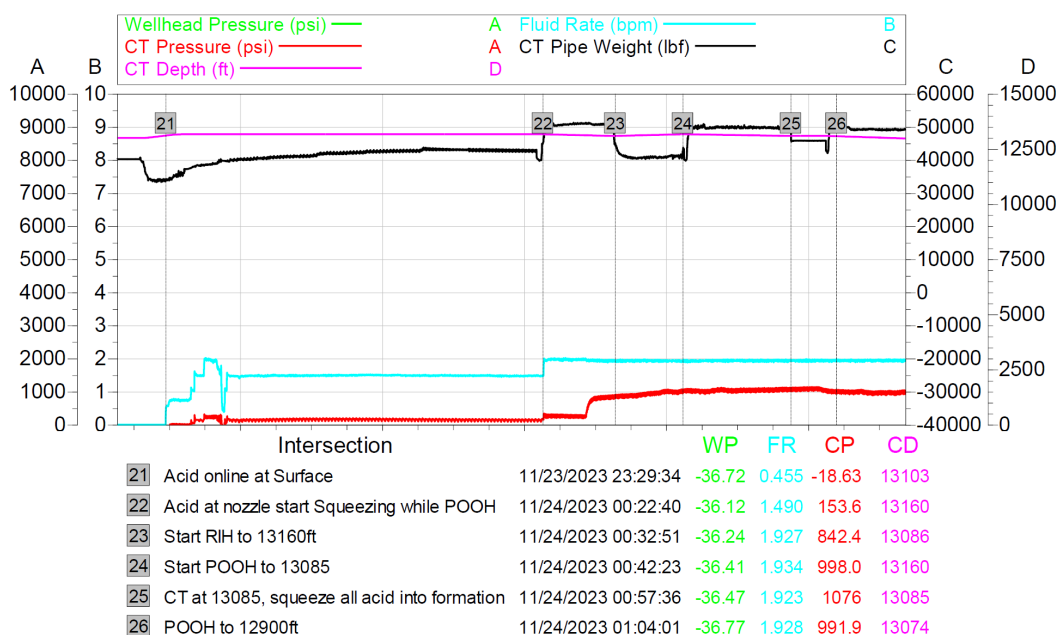


Figure 14—Perforation with 20% HCl acid using FOTFA.



## Stimulation

The well was closed from the choke manifold, and a specific pumping sequence followed. After the perforation wash, wellhead pressure dropped to negative rapidly. Suction behavior remained prominent, and no pressure changes were observed afterward (Fig. 15).

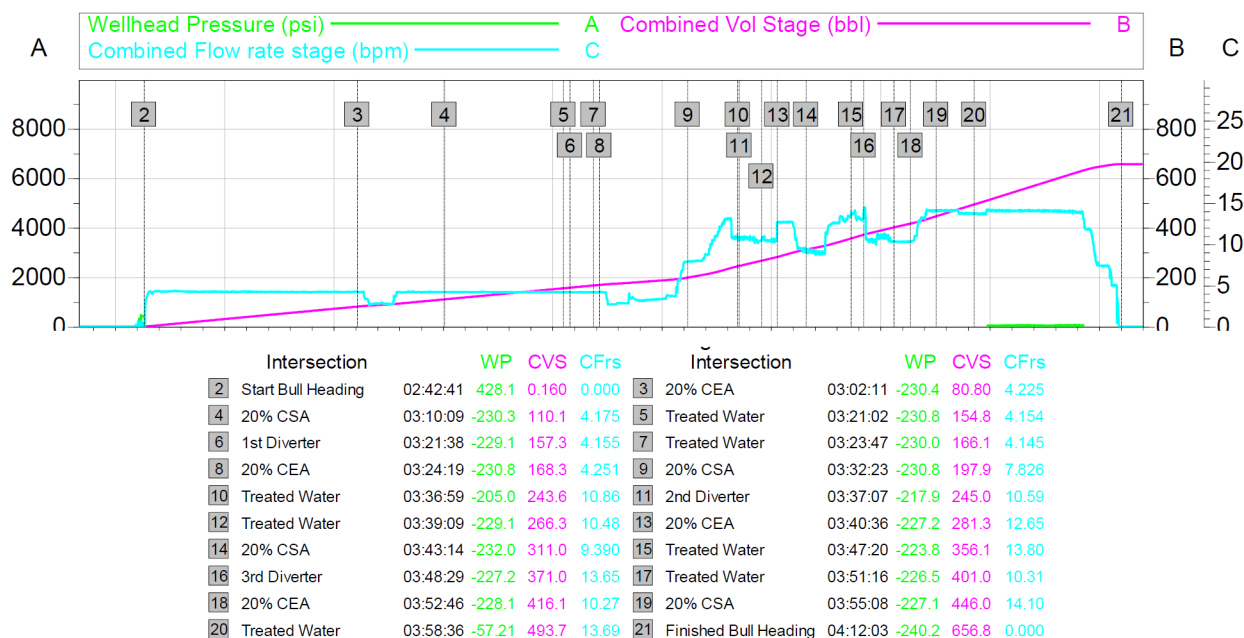


Figure 15—Execution graph showing the diversion indication after each SFAD stage.

The suction behavior indicates that the acid bypassed the damaged zone and established communication between the wellbore and reservoir. A two-hour soaking period followed the matrix stimulation. A nitrogen lift operation was performed from 5,000 ft because of the lack of natural flow. Production reached a level 50% higher than post-acid fracture rates at the same choke size. SFAD application helped prevent a potential increase in water cut, which remained 5% lower than the initial water cut value.

## Results and Conclusion

Efficiency and innovation remain essential to the current oil industry landscape. The sector demands greater output with fewer resources along with operational sustainability enhancements. Stimulation operations require significant resources and can exceed budget limits without proper planning. Diagnosis-driven designs are imperative during stimulation operations to maximize asset value for operators.

This study highlights the importance of diversion in water-producing formations and emphasizes the effectiveness of tailored diversion under fractured permeability and subhydrostatic conditions. The case demonstrated superior results from SFAD with fewer resources compared to other diversion mechanisms.

The operation yielded the following key takeaways:

- Accurate diagnosis of well challenges establish the foundation for stimulation design.
- Acid system, volume, and diversion mechanism selection should align with diagnosed well challenges.
- Fractured-permeability wells, significantly higher than the matrix permeability, often qualify for mechanical diversion methods, such as ball sealers, plugs, and packers. However, these options provide no control of acid entry into water-bearing zones.

- Conventional diverters have advantages and disadvantages related to flowback and degradation requirements. SFAD helps eliminate these concerns through a mechanism that operates independently of pH changes and acid reaction rates.
- Unlike other diversion mechanisms, SFAD functions effectively in a wide range of permeability contrasts from 5 mD to 6 Darcys in both carbonate and sandstone formations.
- Current chemical diversion technologies do not prevent the stimulation of water-bearing zones. SFAD outperforms these options with its intrinsic resistance to acid and other water-based fluids entering such zones.
- Matrix stimulation with 20% HCl acid for the near-wellbore region, DEA for the deeper region, and SFAD for diversion produced superior results for the candidate well. Post-treatment outcomes exceeded those obtained from initial acid fracturing operations.
- The diagnosis-driven design surpassed expectations. The operator achieved 50% higher production compared to initial productivity of the well.

Table 3 summarizes the diverters and their impact on the prevention of water zone stimulations.

**Table 3—Summary of diverters and their impact on water-bearing zones.**

| Diverter Type       | Helps Prevent Water Zone Stimulation? | Comments                                                                                                                                       |
|---------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Ball sealers        | No                                    | Diversion is driven by permeability contrast.                                                                                                  |
| Benzoic flakes      | No                                    |                                                                                                                                                |
| Salt granular       | No                                    |                                                                                                                                                |
| Wax and resins      | No                                    |                                                                                                                                                |
| Gelled acids        | No                                    |                                                                                                                                                |
| High-rate diversion | No                                    | Helps prevent acidizing of the water zone to the wellbore level; water could channel behind the casing/liner.                                  |
| Packers             | No                                    |                                                                                                                                                |
| Plugs               | No                                    |                                                                                                                                                |
| Coiled tubing       | No / Yes                              | Dynamic diversion helps prevent stimulation of the water zone through the delivery of a pinpoint treatment to the intended intervals.          |
| SFAD                | Yes                                   | Reduces water permeability with minimal impact on hydrocarbon permeability; halting access of acid (water phase fluid) in water-bearing zones. |

## Nomenclature

|                  |                                                 |
|------------------|-------------------------------------------------|
| BHA              | bottomhole assembly                             |
| BHT              | bottomhole temperature                          |
| CO <sub>2</sub>  | carbon dioxide                                  |
| CT               | coiled tubing                                   |
| cP               | centipoise                                      |
| DEA              | diesel emulsified acid                          |
| FOTFA            | fluid oscillating tuned frequency and amplitude |
| HCl acid         | hydrochloric acid                               |
| H <sub>2</sub> S | hydrogen sulfide                                |
| ID               | inner diameter                                  |
| LCM              | loss circulation material                       |
| mD               | millidarcy                                      |
| OD               | outer diameter                                  |
| POOH             | pull out of hole                                |

|      |                                |
|------|--------------------------------|
| RIH  | run in hole                    |
| RPM  | relative permeability modifier |
| SFAD | solids-free acid diverter      |
| SG   | specific gravity               |
| TD   | total depth                    |
| TW   | treated water                  |

## References

- Al Sebea, S.H., Abu-Eidah, A.I., Patra, M., et al 2023. Enhanced Productivity Using Real-Time Coiled Tubing Assisted by Temperature Mapping and Integrated Dynamic Diversion Acid Squeeze Technique in a Highly Depleted Reservoir of West Kuwait. Presented at the SPE/ICoTA Well Intervention Conference and Exhibition, The Woodlands, Texas, USA, 21–22 March. <https://doi.org/10.2118/212924-MS>.
- Bilden, D.M., Lacy, L.L., Seiler, F.H., et al 1998. New Water-Soluble Perforation Ball Sealers Provide Enhanced Diversion in Well Completions. Presented at the SPE Annual Technical Conference and Exhibition, New Orleans, Louisiana, USA, 27–30 September. <https://doi.org/10.2118/49099-MS>.
- Brown, R.W., Neill, G.H., and Loper, R.G. 1963. Factors Influencing Optimum Ball Sealer Performance. *J Pet Technol.* **15** (04): 450–454. <https://doi.org/10.2118/553-PA>.
- Eoff, L., Dalrymple, D., and Reddy, B.R. 2004. Development of Associative Polymer Technology for Acid Diversion in Sandstone and Carbonate Lithology. Presented at the SPE/DOE Symposium on Improved Oil Recovery, Tulsa, Oklahoma, USA, 17–21 April. <https://doi.org/10.2118/89413-MS>.
- Eoff, L., Dalrymple, D., and Reddy, B.R. 2005. Development of Associative Polymer Technology for Acid Diversion in Sandstone and Carbonate Lithology. *SPE Prod & Fac.* **20**: 250–256. <https://doi.org/10.2118/89413-PA>.
- Gdanski, R.D. 2005. Recent Advances in Carbonate Stimulation. Paper presented at the International Petroleum Technology Conference, Doha, Qatar, 21–23 November. <https://doi.org/10.2523/IPTC-10693-MS>.
- Ghommam, M., Zhao, W., Dyer, S., et al 2015. Carbonate acidizing: Modeling, analysis, and characterization of wormhole formation and propagation. *J Pet Sci and Eng.* **131**: 18–33. <https://doi.org/10.1016/j.petrol.2015.04.021>.
- Harrison, N.W. 1972. Diverting Agents-History and Application. *J Pet Technol.* **24** (05): 593–598. <https://doi.org/10.2118/3653-PA>.
- Kalfayan, L.J. and Martin, A.N. 2009. The Art and Practice of Acid Placement and Diversion: History, Present State and Future. Presented at the SPE Annual Technical Conference and Exhibition, New Orleans, Louisiana, USA, 4–7 October. <https://doi.org/10.2118/124141-MS>.
- Kleverlaan, M., van Noort, R.H., and Jones, I. 2005. Deployment of Swelling Elastomer Packers in Shell E&P. Presented at the SPE/IADC Drilling Conference, Amsterdam, Netherlands, 23–25 February. <https://doi.org/10.2118/92346-MS>.
- Leguizamón, J., Jaiyeola, A.I., Al Shehhi, H.A.M., et al 2022. Selective Multi-Stage Stimulation in Open-Hole Completion in a Carbonate Formation Using Dynamic Diversion: Case History in UAE. Presented at the SPE Annual Technical Conference and Exhibition, Houston, Texas, USA, 3–5 October. <https://doi.org/10.2118/210423-MS>.
- MaGee, J., Buijse, M.A., and Pongratz, R. 1997. Method for Effective Fluid Diversion when Performing a Matrix Acid Stimulation in Carbonate Formations. Presented at the Middle East Oil Show and Conference, Bahrain, 15–18 March. <https://doi.org/10.2118/37736-MS>.
- McCulloch, D.J., Mann, J., Macmillan, P., et al 2003. Damage Removal in Screened Horizontal Wells. Presented at the SPE/ICoTA Coiled Tubing Conference and Exhibition, Houston, Texas, USA, 8–9 April. <http://dx.doi.org/10.2118/81732-MS>.
- Paccaloni, G. 1995. A New, Effective Matrix Stimulation Diversion Technique. *SPE Prod & Fac.* **10** (03): 151–156. <https://doi.org/10.2118/24781-PA>.
- Singh, J.R. 1985. An Evaluation of Acid Frac/Matrix Stimulation of a Tight Limestone Formation in Exploratory Wells in Kuwait. Presented at the Middle East Oil Technical Conference and Exhibition, Bahrain, 11–14 March. <https://doi.org/10.2118/13751-MS>.
- Talabani, S., Hareland, G., Rajtar, J.M., et al 1997. Optimized Field Matrix Acid Stimulation Parameters in a Multi-Layered Reservoir Using Foam Diversion. Presented at the Latin American and Caribbean Petroleum Engineering Conference, Rio de Janeiro, Brazil, 30 August–3 September. <https://doi.org/10.2118/38958-MS>.
- Thompson, K.E. and Gdansk, R.D. 1993. Laboratory Study Provides Guidelines for Diverting Acid With Foam. *SPE Prod & Fac.* **8** (04): 285–290. <https://doi.org/10.2118/23436-PA>.
- Trif, V.C., Ganea, V.A., and Foidas, I. 2024. Experimental Study of Self-Degrading Diverting Agent Particles Used for Improving Hydraulic Fracture Placement. Presented at the SPE International Conference and Exhibition on Formation Damage Control, Lafayette, Louisiana, USA, 21–23 February. <https://doi.org/10.2118/217860-MS>.

- Vasquez, J., Waltman, B., and Eoff, L. 2010. Field Implementation of a Novel Solids-Free System to Minimize Fluid Loss during Overbalanced Workover Operations. Presented at the SPE EUROPEC/EAGE Annual Conference and Exhibition, Barcelona, Spain, 14–17 June. <http://dx.doi.org/10.2118/130210-MS>.
- Webb, E.D., Schultz, R.L., Howard, R.G. et al 2006. Next-Generation Fluidic Oscillator. Presented at the SPE/ICoTA Coiled Tubing Conference & Exhibition, The Woodlands, Texas, USA, 4–5 April. <http://dx.doi.org/10.2118/99855-MS>.
- Williams, B.B., Gidley, J.L. and Schechter, R.S. 1979. Acidizing Fundamentals. Richardson, TX: Monograph Series, Society of Petroleum Engineers.